

Master Thesis - Outline and Results

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Introduction

- **The lack of legal empirical research:** Empirical legal research is still scarce, at least in Brazil. In the recent decades there were the idea of bringing a quantitative approach to Law, called jurimetrics Loevinger ([1948](#)). Recently, it is becoming more common to find studies based on empirical research as showed by Cabrera De Bonito et al. ([2025](#)).
- **SOTA of legal research:** Those studies, however, still uses little data from lawsuits and judicial precedents. Some of them do more qualitative/observational methods. To do more deep and broad analysis it is necessary to have a high level of resources and big and highly qualified research team. Examples of very good and important empirical research showed the relevant of this staff: Associação Brasileira de Jurimetria ([2019](#)), Costa et al. ([2011](#)), Instituto de Pesquisa Econômica Aplicada (Ipea) & Brasil. Ministério da Justiça e Segurança Pública ([2023](#)).
- **Exploratory legal data analysis:** It is possible to simple analyze metadata of legal cases. Doing this would be useful to exploratory analysis and to formulate hypothesis. Data from lawsuits are also used as some examples shows. In Dornelles ([2021](#)) data extracted from the Brazilian Supreme Court were used to describe the volume of new actions in the Court during Bolsonaro's administration:

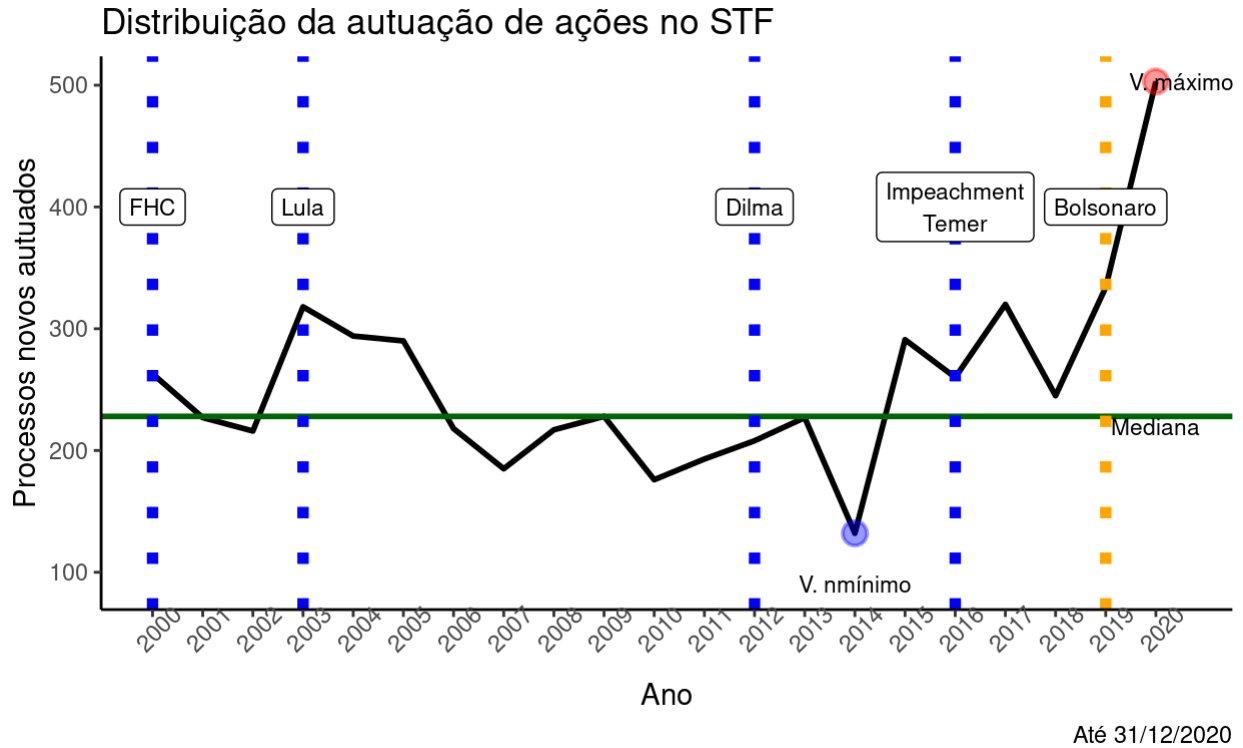


Figure 1: Evolution of new lawsuits each year in Brazilian Supreme Court. Dornelles (2021)

- **Robust research:** However, to do more legal research that goes beyond metadata. And, to do that, it is necessary to analyze complex documents in depth. Those documents are often relatively large and full of legal jargon, meaning that is also necessary that legal specialists do the analysis or, at least, oversee it.
- **Natural Language Processing:** NLP is an important tool to handle text as data and to provide valuable insights from text documents. However, NLP models often are specialized in a single task Vajjala et al. (2020), while, to extract information from such documents it is necessary a handful of tools. Besides that, the traditional NLP models have little ability to understand larger contexts and to follow specific instructions Raschka (2024), so large language models would be a better approach to that. [1].
- **Opportunity:** The advent of powerful language models plus the high level of digitization of the Judiciary branch in Brazil would allow to a new era of empirical research,

both into the academic and the practitioners perspective. It would be possible that a single researcher study how administrative misconduct law cases were decided and the last few years. Also, it would be possible to a Public Defender study Local Court's decisions in order to establish better strategy to new cases.

- **This research:** The aim is to understand if and how large language models could be used to leverage legal research by processing whole judicial decisions and extracting the features of interest. Moreover, the interest is about understanding the feasibility to use open-source large language model in real-world settings.
- **Open-source models:** In academia, in the public service and similar use cases, privacy and costs are a limitation. Although the terms of use of commercial LLMs might suggest a relative degree of confidence about data protection, once inserted in the API there is no real control over it. Also, the commercial models might be expensive (although cheaper than the fair price of human labor to do a similar task), which might be a block to its deployment.¹

Literature Review

- **Brazilian legal research landscape:** Alves da Silva (2022), Cabrera De Bonito et al. (2025)
- **LLMs applied to legal field:** Junior et al. (2024), Katz et al. (2023), Bertalan & Ruiz (n.d.)
- **LLMs in Portuguese:** Niklaus et al. (2023), Almeida et al. (2024), Pires et al. (2023)
- **Specialization in Legal Tasks:** This is the paper that most gave actionable inputs to this research: Dominguez-Olmedo et al. (2024)

¹Although there is some debate around Small Language Models and Large Language Models, this distinction is not being made here. See also Chen (2025)

Research Question

Can open-source large language models be deployed in the Brazilian legal context to accurately extract information, recognize entities, and summarize complex legal documents?

Data

- **Lacerda Dataset:** Lacerda e Silva (2021) did a comprehensive empirical analysis on over 400 legal cases related to drug traffic. She gathered those cases in São Paulo's State Court of Justice (TJSP) in a sample covering between 2017 to 2018 and she annotated and labelled about 35 features of each case manually. I received the data from the researcher and adapted the dataset to this research.

Some of the original features also referred to other documents rather than the judicial decision and, therefore, they were excluded (since it would not be reasonable that the model inferred that). Also, features that were related to biological data were excluded due ethical reasons.

Also, I have tried to access Instituto de Pesquisa Econômica Aplicada (Ipea) & Brasil. Ministério da Justiça e Segurança Pública (2023) micro data to build a golden dataset since they have about 3.5k data points labelled. I have filled a freedom of information request and, although it was possible to access the [general dataset](#), they have not provided the lawsuit identifier, so it was not possible to match each row in their dataset to a legal case and, therefore, to a specific document.

- **Judicial Decisions:** The documents relative to each decision case analyzed by the Lacerda Dataset were extracted automatically from the TJSP website. Since there is no API to that, the process was done using Filho & Trecenti (2020) package called `{tjsp}`. It gathered metadata of each case and the decision in string format in a R dataframe. Some cases were collected by its .pdf files and read automatically.
- **Final dataset:** After the cleaning process, some duplication were excluded and some

documents were not collected, the final result was a dataset with 258 cases. Also, the final dataset contained 50 columns: it was due some categorical features were transformed in boolean. For instance: the column *DenDrogas* contained information about the most common articles of law that were in the accusation. It was transformed in the columns: *denuncia_art_33*, *denuncia_art_34* and *denuncia_art_35*.

Theoretical Framework

- **Natural Language Processing**
- **LLMs**
- **Fine tuning:** Using Dominguez-Olmedo et al. (2024) strategy.

Models

We used 12 models in this research. Models from OpenAI were considered commercials and the others open source. Also, OpenAI do not disclose their architectures, so it is not possible to know beforehand their number of parameters.

model	family	parameters	type
GPT 4.1	OpenAI	unknown	commercial
GPT 4.1-mini	OpenAI	unknown	commercial
GPT 4.1-nano	OpenAI	unknown	commercial
GPT 4o	OpenAI	unknown	commercial
GPT 4o-mini	OpenAI	unknown	commercial
o3-mini	OpenAI	unknown	commercial
Gemma 3	Gemma	27 B	open-source
Gemma 3	Gemma	12 B	open-source
Gemma 3	Gemma	1 B	open-source

Llama 3.1	Llama	8 B	open-source
Llama 3.2	Llama	3 B	open-source
Phi 4	Phi	14 B	open-source
Lawma 8B	Lawma	8 B	open-source
Tucano	Tucano	2.4 B	open-source

Methods

- etl
- prompt engeneenering, build schema
- generation in OpenAI's models
- evaluation pipeline
- generation with commercial models
 - window context: rope scaling, yarn, rag, chuncking
 - structured output
 - base model generation
- finetune commercial and open source models
 - OpenAI's API
 - Unsloth
- final evaluation

Results

After 17 experiments², it was possible to gather the metrics above regarding accuracy. By accuracy we are considering the proportion of exact answers regarding the total answers. Bellow, we can see the performance of the base models, before fine tuning.

²Ignoring validation tests and the experiences to try to overcome the limitations regarding context windows. Also, in this count there were the fine tune.

family	type	model	parameters	accuracy
Gemma	open-source	Gemma 3	27 B	0.8200
OpenAI	commercial	GPT 4o	unknown	0.7878
OpenAI	commercial	o3-mini	unknown	0.7638
Gemma	open-source	Gemma 3	12 B	0.7627
OpenAI	commercial	GPT 4.1	unknown	0.7545
Phi	open-source	Phi 4	14 B	0.7474
Llama	open-source	Llama 3.1	8 B	0.7354
OpenAI	commercial	GPT 4o-mini	unknown	0.7114
OpenAI	commercial	GPT 4.1-mini	unknown	0.6967
OpenAI	commercial	GPT 4.1-nano	unknown	0.6590
Llama	open-source	Llama 3.2	3 B	0.5423
Gemma	open-source	Gemma 3	1 B	0.0000

The model Gemma 3-1B achieved 0.0 accuracy due their lack of abilities to yield structured JSON outputs.

The model that best performed in a zero-shot approach was the Gemma 3-27B parameters, which, surprisingly, also beat the benchmark commercial models, even GPT-4o. Also, the Gemma 3 family seems to be very prone to yield accurate results, since the lighter version with 12B parameter also performed very well and was better than the newly released GPT-4.1. The model Phi-4 performed also very well, also betting OpenAI’s smaller models. Regarding the Llama family, the 3.1 version performed better than 4o-mini and the smaller GPT 4.1-mini and GPT 4.1-nano. On the other hand, Llama 3.2 had a bad performance.

Trying to understand a more fine-grained analysis, it was possible to break the results by the kind of tasks. The *numeric* tasks are those related to extract and calculate the amount of

drugs or the prison time in months: those tasks involved not only recognizing the numbers (entities) but also to ‘calculate’ the final value. The *boolean* tasks are generally in the realm of question answering in a TRUE/FALSE/None style (for instance, if the art. 33 was the basis of the sentence or if there were confession by the defendant). The *categorical* tasks are those closely related to classification tasks, such as what was the outcome of the case (conviction, acquittal, etc.). The *open textual* tasks are more likely to be similar to named entity recognition: in this case, the name of the Judge and the defendant.

	Accuracy by model and experiment				
	overall_accuracy	numeric	boolean	categorical	open_textual
GPT 4o-mini	0.93	0.88	0.97	0.77	0.96
Phi 4	0.93	0.85	0.94	0.92	0.95
Gemma 3	0.82	0.82	0.81	0.82	0.95
GPT 4o	0.79	0.88	0.75	0.87	0.96
Llama 3.2	0.78	0.74	0.78	0.81	0.85
o3-mini	0.76	0.86	0.75	0.67	0.94
Gemma 3	0.76	0.74	0.77	0.66	0.94
GPT 4.1	0.75	0.88	0.71	0.86	0.96
Phi 4	0.75	0.76	0.73	0.78	0.87
Llama 3.1	0.74	0.73	0.73	0.68	0.92
GPT 4o-mini	0.71	0.77	0.67	0.81	0.95
GPT 4.1-mini	0.70	0.83	0.64	0.82	0.91
GPT 4.1-nano	0.66	0.80	0.61	0.68	0.96
Llama 3.2	0.54	0.46	0.58	0.33	0.64

We can see that the *open textual* tasks have a very high level of performance in nearly all models, while the boolean and categorical tasks were more diverse performance-wise. We can see also some differences in numeric tasks, but in general they all performed very well.

Those results suggested that keep working with open source models were a promising strategy. So fine tuning was performed with the models OpenAI 4o-mini (which is the recommended model by OpenAI), Llama 3.2 (3B) and Phi4. There were no resources available to perform fine tune on other models. After the fine tuning, we could see a big improvement in the results.

The process of fine tuning, even with a small dataset with 186 cases, highly improved the

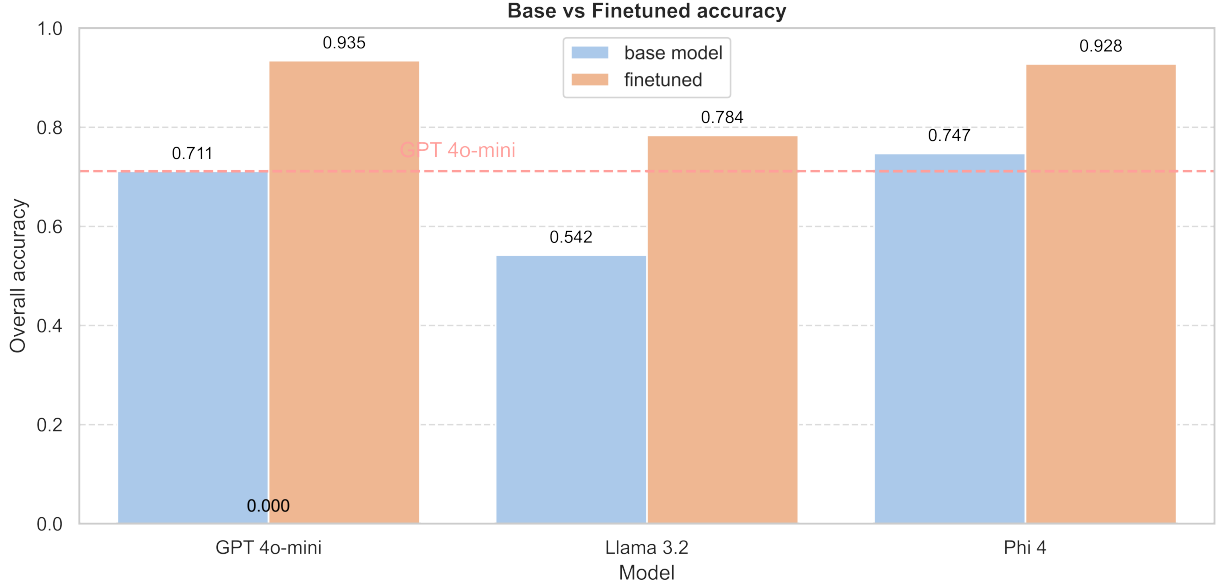


Figure 2: Models after fine tuning

performance of the models, confirming the ‘power of specialization’ as suggested by (?). As (?) showed, model Phi4 tied with GPT 4o-mini and the much smaller Llama 3.2 (3B) learnt very easily in the process, even both models being trained with the same 10 epochs. It suggests that with a richer dataset and/or more epochs it would be possible to reach or even surpass GPT 4o-mini performance.

metric	GPT 4o-mini	Llama 3.2	Phi 4
Δ accuracy	0.22	0.24	0.18
% accuracy	31.37	44.57	24.16
Δ boolean	0.29	0.20	0.21
% boolean	43.56	34.54	28.63
Δ categorical	-0.04	0.48	0.14
% categorical	-4.46	145.31	17.76
Δ numeric	0.11	0.28	0.09
% numeric	14.44	62.15	11.17

Δ open textual	0.01	0.21	0.08
% open textual	1.35	33.00	8.82

Table 3: Fine tuned models: change in accuracy (absolute value Δ and relative value %)

As Table 3 shows, Llama 3.2(3B) had the highest change in its performance, specially with the categorical tasks. Specifically with this kind of tasks, we could see a degradation of performance of GPT 4o-mini. Phi4, although had the second-best performance, learnt relatively less than Llama which might suggest a ceiling. Assuming that, in those architectures, the number of parameters is the critical factor, maybe less parameters than 14B and more than 3B would be ideal.

Policy Recommendations

- Academia: disclose data from empirical legal research as much as possible, as it would be a valuable resource to fine tune models. If IPEA released the data from Instituto de Pesquisa Econômica Aplicada (Ipea) & Brasil. Ministério da Justiça e Segurança Pública (2023), would it be possible to achieve even better results.
- Open source community: keep the standard of at least 128k context window and develop light open source models. Wolf et al. (2020) is a very important framework to training and inference. Daniel Han & team (2023) was valuable to efficiently fine tune models efficiently.
- Judiciary branch: make sure data and documents from lawsuits are available digitally for research, keeping high standards of privacy. It would be feasible through anonymization.

³There is the [mini version of Phi-4](#), however it was not possible to run or fine tune it with the actual computational resources.

- Institutions interested in empirical legal research: the balance of easiness and performance are probably better with OpenAI's API. But open source models showed to be powerful to this tasks and might be deployed to accelerate empirical research in Law.
- Law schools: invest in data literacy to legal students and practitioners.

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Appendix I - Lacerda's Dataset

Name	lacerda_example
Number of rows	420
Number of columns	35
Column type frequency:	
character	35
Group variables	None

Table 4: Data summary

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
Processo	0	1	20	26	0	352	0
DtDisp	0	1	5	10	0	210	0
Juiz	0	1	13	49	0	79	0
Vara	0	1	16	17	0	32	0
Nome	0	1	11	39	0	419	0
Local	0	1	7	19	0	10	0
Sexo	0	1	8	9	0	2	0
DN	0	1	5	28	0	411	0
Cutis	0	1	5	7	0	5	0
Cabelo	0	1	4	16	0	8	0
Objetos	0	1	4	159	0	261	0
Maconha	0	1	1	41	0	255	0
Cocaína	0	1	1	38	0	274	0
Crack	1	1	1	40	0	138	0
Ecstasy	0	1	1	24	0	12	0
LSD	0	1	1	18	0	6	0
Outras	0	1	1	61	0	33	0
DenDrog	0	1	7	25	0	4	0
DenOutros	0	1	3	51	0	13	0
Aspectos	0	1	6	122	0	156	0
Sentença	0	1	10	23	0	5	0
ResDrogas	0	1	2	25	0	6	0
ResOutros	0	1	2	45	0	16	0
AvalNegPB	0	1	2	50	0	28	0
PenaBase	0	1	2	12	0	18	0

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
Agravantes33	0	1	2	12	0	3	0
Atenuantes33	0	1	2	21	0	5	0
Aumento33	0	1	2	26	0	8	0
§4º	0	1	2	13	0	5	0
Pena33	0	1	2	10	0	46	0
PenaDrogas	0	1	2	5	0	7	0
PenaOutros	0	1	2	9	0	15	0
TotPen	0	1	2	11	0	59	0
Substituição da pena	0	1	2	3	0	3	0
Regime inicial	0	1	2	10	0	4	0

Appendix II - Prompt

Appendix III - Fine tune performance

Appendix IV - Plots and tables

Appendix V - Acknowledgments